



QT-WEBKIT:

A QUICK WAY TO DEVELOP A DESKTOP OR MOBILE APP

Qt-WebKit, a major engine that can render Web pages and execute JavaScript code, is the answer to the developer's prayer. Let's take a look at a few examples that will aid developers in making better use of this engine.

This article is for Qt developers. It is assumed that the intended audience is aware of the famous Signals and Slots mechanisms of Qt. Creating an HTML page is very quick compared to any other way of designing a GUI. An HTML page is nothing but a fancy page that doesn't have any logic in its build. With the amalgamation of JavaScript, however, the HTML page builds in some intelligence. As everything cannot be collated in JavaScript, we need a back-end for it. Qt provides a way to mingle (HTML+Java) with C++. Thus, you can call the C++ methods through JavaScripts and vice-versa. This is possible by using the Qt-WebKit framework. The applications developed in Qt are not just limited to various desktop platforms. They are even ported over several mobile platforms. Thus, you can design your apps that can just fit into the Windows, iOS and Android worlds, seamlessly.

What is Qt-WebKit?

In simple words, Qt-WebKit is the Web-browsing module of Qt. It can be used to display live content from the Internet as well as local HTML files.

Programming paradigm

In Qt-WebKit, the base class is known as QWebView. The sub-class of QWebView is QWebViewPage, and a further sub-class is QWebFrame. This is useful while adding the desired class object to the JavaScript window object. In short, this class object will be visible to JavaScript once it is added to the JavaScript window object. However, JavaScript can invoke only the public *Q_INVOKABLE* methods. The *Q_INVOKABLE* restriction was introduced to make the applications being developed using Qt even more secure.

Q_INVOKABLE

This is a macro that is similar to Slot, except that it has a return type. Thus, we will prefix *Q_INVOKABLE* to the methods that can be called by the JavaScript. The advantage here is that we can have a return type with *Q_INVOKABLE*, as compared to Slot.

Developing a sample HTML page with JavaScript intelligence

Here is a sample form in HTML-JavaScript that will allow us to multiply any two given numbers. However, the logic of multiplication should reside in the C++ method only.

```
<html>
<head>
<script>
function Multiply()
{
    /** MultOfNumbers a C++ Invokable method */
    var result = myoperations.MultOfNumbers(document.forms["DEMO_FORM"]["Multiplicant_A"].value, document.forms["DEMO_FORM"]["Multiplicant_B"].value);
    document.getElementById("answer").value = result;
}
</script>
</head>
<body>
<form name="DEMO_FORM">
    Multiplicant A: <input type="number"
    name="Multiplicant_A"><br>
    Multiplicant B: <input type="number"
```